Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

Contract All Courses |

Statistics

- ### STAT 0996: Support for Elementary Statistics

1 Credit Hours

Corequisite: STAT 1401

Description
This Learning Support course provides co-requisite support for students enrolled in STAT 1401 - Elementary Statistics. Topics will parallel topics being studied in STAT 1401 and the course will provide support for the essential skills needed to be successful in STAT 1401. Taken with STAT 1401, topics to be covered will include descriptive statistics, probability theory, confidence intervals, hypothesis testing, and other selected statistics topics.

- ### STAT 1401: Elementary Statistics

3 Credit Hours

Description
This is a non-calculus based introduction to statistics. Course content includes descriptive statistics, probability theory, confidence intervals, hypothesis testing, and other selected statistical topics.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Use statistical vocabulary and notation appropriately.
2. Identify appropriate methods for collecting data.

3. Distinguish the difference between qualitative and quantitative data and recognize when each is appropriate.
4. Describe and graphically represent statistical data.
5. Correctly interpret statistical graphical displays.
6. Identify measures of center and variation and use them appropriately to describe distributions.
7. Compute basic probabilities and correctly use computations for application.
8. Build confidence intervals to estimate population parameters such as means and proportions from statistical data.
9. Perform hypothesis tests for population parameters and appropriately interpret the results.
10. Successfully use technology to describe, analyze, and perform inferential statistics.
11. Apply statistical concepts to real-life scenarios.

- ## STAT 2332: Probability and Data Analysis

3 Credit Hours

Prerequisite: MATH 1190 or MATH 1179

Description

This course is a foundational, calculus-based introduction to statistics and probability. The following conceptual themes will be developed through the process of statistical investigation: exploratory data analysis (univariate and bivariate), fundamentals of experiment design and sampling, planning and conducting a study, exploring random phenomenon using probability and simulation, and the fundamentals of statistical inference. Technology is integrated into each theme, and the statistical software package used will be chosen by the instructor.

- ## STAT 3120: Statistical Methods I

3 Credit Hours

Prerequisite: STAT 1401 or DATA 1501 or STAT 2332 or MATH 3332

Description

This course is designed to provide students with a foundation in statistical methods, including confidence intervals for population parameters, correlation, simple linear regression and hypothesis testing (F and T-tests for regression, chi-square for independence, 2 group and paired sample T-tests). These concepts are taught with heavy emphasis on statistical coding software and real-world datasets from a variety of disciplines.

- ## STAT 3125: Biostatistics

3 Credit Hours

Prerequisite: BIOL 1107 or BIOL 1108 or CHEM 1212 or permission of the instructor

Description

In this course students use descriptive statistics and visual displays to describe biological and medical data. They perform and analyze results of statistical analyses which may include confidence intervals, correlation, linear regression, odds/risk ratios, and hypothesis testing (Chi-square for independence, 2 group and paired sample t-tests). Analyses are performed using the statistical software R.

- ## STAT 3130: Statistical Methods II

3 Credit Hours

Prerequisite: DATA 3010 and (STAT 3120 or STAT 3125 or STAT 2332 or PSYC 3000)

Description

Students continue to build their foundation in statistical methods in this course. They will conduct non-parametric methods (Wilcoxon Signed Rank, Rank Sum, and Kruskal Wallis tests), ANOVA and multiple regression. These concepts are taught with heavy emphasis on statistical coding software and real-world datasets.

- ## STAT 4025: Clinical Trial Design

3 Credit Hours

Prerequisite: STAT 3125 or STAT 3120

Description

The course introduces students to statistical concepts used to design clinical trials, or randomized studies of humans. Students will be able to design, conduct, and analyze clinical trials in the format required by the Food and Drug Administration. The topics include endpoint definition, sources of bias, randomization schemes, types of blindness, phases of clinical studies (I-IV), hypothesis formation, sample size determination, patient recruitment, adverse events, and protocol development.

- ## STAT 4120: Applied Experimental Design

3 Credit Hours

Prerequisite: STAT 3130

Description

Methods for constructing and analyzing designed experiments are the focus of this course. The concepts of experimental unit, randomization, blocking, replication, error reduction, and treatment structure are introduced. The design and analysis of completely randomized, randomized complete block, incomplete block, Latin square, split-plot, repeated measures, factorial, and fractional factorial designs will be covered. Statistical software will be utilized.

- ## STAT 4125: Analysis of Human Studies

3 Credit Hours

Prerequisite: STAT 3130

Description

Real-world human-subject data will be used as students conduct official clinical research in a secure computer lab. Students will complete required institutionally approved training for research on human subjects and information privacy and security. Using statistical software, students will apply statistical analysis and modeling techniques to answer questions posed by clinicians. Students will document research for reproducibility and potential publication, as well as present results in various forms.

- ## STAT 4210: Applied Regression Analysis

3 Credit Hours

Prerequisite: STAT 3130

Description

This course is designed to provide students with various regression procedural methods, including simple linear and multiple regression models. Students will diagnose multicollinearity, identify outliers and influential observations, and assess assumptions to create and validate models. Other topics will include parameters inferences, variable transformations, and qualitative predictors.

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